AP CALCULUS AB - UNIT 3

Guided Notes 3.1: The Chain Rule

Name: ______________ **Class:** ______________ **Date:** ______________

Learning Objectives and Essential Question

Essential question: How can we recognize composite functions and identify their inner and outer functions?

- Recognize composite functions and identify their inner and outer functions.
- Differentiate composite functions using the Chain Rule in prime and Leibniz notation.
- Combine the Chain Rule with power, trigonometric, exponential, logarithmic, product, and quotient rules.
- Use tables, graphs, and contextual information to evaluate derivatives of compositions.

Key Knowledge, Vocabulary, and Notation

Composite function — A function of the form $f(g(x))$; the output of g becomes the input of f.

Chain Rule — $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$.

Leibniz form — If $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$.

Nested composition — Differentiate from the outside inward and multiply by every inner derivative.

Evaluation from data — $(f \circ g)'(a) = f'(g(a))g'(a)$; locate $g(a)$ first.

Units — Rates multiply so intermediate units cancel, leaving output units per input unit.

Concept Development and Strategy

1. Mark the innermost expression before differentiating. Rewrite radicals and reciprocals as powers when useful.
2. Differentiate the outside while leaving the inside unchanged, then multiply by the derivative of the inside.
3. For several layers, repeat the process until the derivative reaches the original independent variable.
4. When product or quotient structure is present, preserve that structure and apply the Chain Rule inside its factors.
5. Do not write $f'(x)g'(x)$ for the derivative of $f(g(x))$; the outer derivative is evaluated at $g(x)$.

Shared Representation / Data

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	4	−2	2	3
2	−1	5	3	−1
3	6	4	1	2

Worked Example: Power outside

Differentiate $y = (3x^2 - 5x + 1)^4$.

Answer and Reasoning

Let $u = 3x^2 - 5x + 1$. Then $\dfrac{dy}{dx} = 4u^3\dfrac{du}{dx} = 4(3x^2 - 5x + 1)^3(6x - 5)$.

Worked Example: Three nested layers

Differentiate $y = e^{\sin(2x)}$.

Answer and Reasoning

Differentiate exponential, then sine, then $2x$: $y' = 2e^{\sin(2x)}\cos(2x)$.

Worked Example: Product and Chain Rules

Differentiate $y = x^2 \ln(1 + x^3)$.

Answer and Reasoning

$y' = 2x\ln(1 + x^3) + x^2\dfrac{3x^2}{1 + x^3} = 2x\ln(1 + x^3) + \dfrac{3x^4}{1 + x^3}$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Identify inner and outer functions for $y = \sqrt{5x^2 + 1}$.

2. Differentiate $y = (2x - 7)^5$.

3. Differentiate $y = e^{x^2 - 3x}$.

4. Differentiate $y = (1 + \tan x)^6$.

5. Differentiate $y = x^3 \cos(x^2)$.

6. Find all inputs where the derivative of $y = e^{x^2 - 4x}$ is zero.

7. If differentiable f is even, prove that f' is odd.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. If $y = (1 - 3x^2)^5$, then y' is

(A) $5(1 - 3x^2)^4$
(B) $-30x(1 - 3x^2)^4$
(C) $-6x(1 - 3x^2)^5$
(D) $5(-6x)^4$

2. If $h = f \circ g$, which equals $h'(a)$?

(A) $f'(a)g'(a)$
(B) $f'(g(a)) + g'(a)$
(C) $f'(g(a))g'(a)$
(D) $f(g'(a))$

3. At $x = 0.8$, the derivative of $F(x) = e^{\sin(x^2)}$ is closest to

(A) 1.36
(B) 1.88
(C) 2.33
(D) 3.41

Common Errors and AP Precision

- Mark every layer or every occurrence of y before differentiating; omitted inner derivatives are the most common source of error.
- For inverse derivatives, find the preimage first. For implicit relations, verify the point lies on the original curve.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Let $H(x) = g(g(x))$. If $g(1) = 2$, $g'(1) = -3$, and $g'(2) = 4$, find $H'(1)$.